Quantum mechanics

Chapter II Algebraic Formulation of Quantum Mechanics



Linear Vector Spaces



Linear Vector Spaces

Let us start by defining a linear vector space and listing out its properties. In general, a linear vector space consists of a set of elements ψ, φ, χ,... (called vectors) and a set of numbers a, b, c, . . . (called scalars), a set of rules each for the addition and multiplication of vectors.

Definition: A linear vector space V is a set of elements ψ, φ, χ,... called vectors, for which the following properties hold:

1. V is closed under addition. This means that if two vectors ψ and φ belong to V then their sum, written as $\psi + \varphi$, also belongs to V .

2. A vector ψ can be multiplied by a scalar a to yield a new, well-defined vector $a\psi$ that belongs to V ,

3. The addition of vectors is commutative, that is, $\psi+\varphi = \varphi+\psi$.

4. The addition of vectors is associative, that is, $\psi+(\varphi+\chi) = (\psi+\varphi)+\chi$.

5. There exists a unique element called 0 that satisfies $\psi + 0 = \psi$ for every element $\psi \in V$.

6. There exists an identity element, E, in V such that $E\psi = \psi$ for every element $\psi \in V$.

7. The multiplication of a vector by scalars is associative, that is, $(ab)\psi = a(b\psi)$.

8. The multiplication of a vector by a scalar is linear, that is, $a(\psi+\varphi) = a\psi + a\varphi$, $\psi(a+b) = a\psi + b\psi$.

9. For each ψ in V , there exists a unique additive inverse $(-\psi)$ such that $\psi+(-\psi) = 0$.

If the vectors and the scalars associated with a given vector space are real, we say that we are working with a real vector space. On the other hand, if the vectors and the scalars are complex, then we say that we are working with a complex vector space.

Linear Vector Spaces

Linear independence of vectors: Consider a set of n vectors, $\{\phi_1, \phi_2, \phi_3, \ldots, \phi_n\}$, and their linear combination $a_1\phi_1 + a_2\phi_2 + a_3\phi_3 + \ldots + a_n\phi_n$, where a_j= 1,2,3,...,n are all constants. The vectors of this set are said to be linearly independent if the equation

$$a_1\phi_1 + a_2\phi_2 + a_3\phi_3 + \ldots + a_n\phi_n = 0$$

Consider hold only if $a_1 = a_2 = \ldots = a_n = 0$. If this condition is not met, we say that the set is linearly dependent.

Note that if a set of vectors is linearly dependent, one of the vectors can be expressed as a linear combination of the others. For instance, assume that

$$a\psi + b\phi + \ldots + c\chi = 0,$$

where not all of the scalars are zero. Then one of the vectors can be expressed in terms of the other vectors as follows. Let a be non-zero. Then, we have

$$\psi = p\,\phi + \ldots + q\,\chi,$$

and

$$p = -\frac{b}{a}, \ldots, q = -\frac{c}{a}.$$

Linear Vector Spaces

A linear vector space, V , is said to have dimension n, if the maximum number of linearly independent vectors in V equals n.

If this number n is finite, the linear vector space is called finite. On the other hand, if it is possible to find any number (as large as possible) of linearly independent vectors in it, then it is called infinite.

Basis: Any set of n linearly independent vectors, $\{\phi_i\}$, i = 1, 2, 3,...,n, belonging to the n-dimensional linear vector space, V , is called its basis. The elements, ϕ_1 , ϕ_2 , ϕ_3 , . . ., of this set are called the basis vectors.

Moreover, a basis is said to be complete if it spans the entire space; that is, there is no need to introduce any additional basis vector.

It also means that every vector ψ of a linear vector space V, with a complete basis, can be written as a unique linear combination of the basis vectors:

$$\psi = c_1\phi_1 + c_2\phi_1 + c_3\phi_3 + ... + c_n\phi_n,$$

where the expansion coefficients c_i, i = 1, 2, 3, . . . , n are called the components of the vector ψ in the basis $\{\phi_i\}$.

Hilbert Space

In quantum mechanics, the linear vector spaces are, as a rule, infinite-dimensional. The so-called Hilbert space plays an exceptional role among all the infinite-dimensional linear vector spaces.

A Hilbert space is equipped with an inner product that is essentially positive and allows to introduce metric relationship among various quantities.

In this sense, a Hilbert space is a natural generalization of Euclidean spaces to infinite-dimensional spaces.

Hilbert Space

A Hilbert space H is a collection of vectors, ψ, φ, χ,... and scalars, a, b, c,... that satisfies the following properties.

1. H is an infinite-dimensional linear vector space, that is, it has infinite dimensions and possesses all the properties of a linear vector space discussed earlier.

2. There exists in H a real inner product which is finite and satisfies all the aforementioned properties.

3. H is separable. $\|\psi - \psi_n\| < \varepsilon.$

4. H is complete. $\lim_{n,m\to\infty} \|\psi_n - \psi_m\| = 0, \qquad \lim_{n\to\infty} \|\psi - \psi_n\| = 0.$

Square-Integrable Functions

In the case of function spaces, a "vector" element is given by a *complex function* and the *scalar product* by *integrals*. That is, the scalar product of two functions $\psi(x)$ and $\phi(x)$ is given by

$$(\psi, \phi) = \int \psi^*(x)\phi(x)\, dx.$$

If this integral *diverges*, the scalar product *does not exist*. As a result, if we want the function space to possess a scalar product, we must select only those functions for which (ψ,ϕ) is *finite*. In particular, a function $\psi(x)$ is said to be *square integrable* if the scalar product of ψ with itself,

$$(\psi, \psi) = \int |\psi(x)|^2\, dx,$$

is finite

Dirac Notation

We have already stated that vector spaces in quantum mechanics are complex. Therefore, we assume the elements of our n-dimensional linear vector space to be complex. We also assume the vector space to have a fixed basis $\{\phi_i\}$, i= 1, 2, 3,...,n.

Dirac notation: We introduce the notation $|\psi\rangle$ for a vector ψ belonging to an n-dimensional linear vector space V, and we call it a ket vector or simply a ket.

In fixed basis $\{\phi_i\}$, i = 1, 2, 3,..., n, a ket will be characterized by its complex components ψ_i, i = 1, 2, 3,...,n.

It is convenient to arrange these components in to a column vector and write a ket as a column vector:

$$|\psi\rangle = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ . \\ . \\ . \\ \psi_n \end{pmatrix}.$$

Dual vector: The familiar notion of a "scalar product" is incorporated by introducing a dual vector, written as $\langle\psi|$, for each of the vectors, $|\psi\rangle$, of V.

In Dirac's language, it is called a bra vector. The bra $\langle\psi|$ dual to a ket $|\psi\rangle$ is constructed by transposing the ket (that is, we write it as a row vector) followed by complex conjugation. In other words:

$$\text{If} \quad |\psi\rangle = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ . \\ . \\ . \\ \psi_n \end{pmatrix}, \quad \text{then} \quad \langle\psi| = \begin{pmatrix} \psi_1^* & \psi_2^* & \psi_3^* & . & . & . & \psi_n^* \end{pmatrix}.$$

This method of complex conjugation is known as hermitian conjugation or dagger † operation: $\langle\psi| = (|\psi\rangle)^\dagger$.

There is a one-to-one correspondence between bras (constructed in this manner) and kets, that is, for a given ket $|\psi\rangle$, there is a unique bra $\langle\psi|$. In addition, the following relations hold good:

(a) If $|\lambda\rangle = \alpha|\mu\rangle$, then $\langle\lambda| = \alpha^*\langle\mu|$.

(b) If $|\lambda\rangle = |\alpha\mu\rangle + \beta|\nu\rangle$, then $\langle\lambda| = \alpha^*\langle\mu| + \beta^*\langle\nu|$.

The set of bras, dual to the kets of V , also forms a linear vector space, which is called the dual (to V) vector space. It is denoted as V*.

The inner (or, scalar) product: The inner product (also called the scalar product) of two vectors $|\psi\rangle$ and $|\phi\rangle$ (written as $\langle\phi|\psi\rangle$) is defined by the following expression:

$$\langle\phi|\psi\rangle = \begin{pmatrix} \phi_1^* & \phi_2^* & \phi_3^* & . & . & . & \phi_n^* \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ . \\ . \\ . \\ \psi_n \end{pmatrix}$$

$$= (\phi_1^*\psi_1 + \phi_2^*\psi_2 + \phi_3^*\psi_3 + ... + \phi_n^*\psi_n) = \sum_{i=1}^{n} \phi_i^*\psi_i.$$

We call $\langle\phi|\psi\rangle$ a 'bracket'.

$$\langle\phi \mid \psi\rangle = \int \phi^*(\vec{r}, t)\,\psi(\vec{r}, t)\,d^3r.$$

Properties of the inner product:

$$\langle\lambda|\psi\rangle = \langle\psi|\lambda\rangle^* = \langle\psi|\alpha\mu\rangle^* = (\alpha\langle\psi|\mu\rangle)^* = \alpha^*\langle\psi|\mu\rangle^* = \alpha^*\langle\mu|\psi\rangle.$$

$$\langle\psi|(\alpha|\phi\rangle + \beta|\omega\rangle) = \alpha\langle\psi|\phi\rangle + \beta\langle\phi|\omega\rangle,$$

$$(\langle\alpha\psi| + \langle\beta\omega|)|\phi\rangle = \alpha^*\langle\psi|\phi\rangle + \beta^*\langle\omega|\phi\rangle,$$

$$\langle\psi|\psi\rangle \geq 0.$$

If the inner product between two vectors is zero, $\langle\phi|\psi\rangle = 0$, we say that the vectors are orthogonal.

Norm of a vector: The square root of the inner product of a vector with itself is called the norm, and is written as:

$$\|\psi\| = \sqrt{\langle\psi|\psi\rangle}.$$

A vector $|\psi\rangle$ is said to be normalized if its norm is equal to 1:

$$\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = 1.$$

Orthonormal and complete basis: An orthonormal basis consists of the basis vectors $\{|\phi_i\rangle\}$, i = 1, 2, 3, . . . , n, which have a unit norm and are pairwise orthogonal:

$$\langle\phi_i|\phi_j\rangle = \delta_{ij}, \quad \|\phi_i\| = \sqrt{\langle\phi_i|\phi_i\rangle} = 1,$$

Let us first assume the basis to be discrete. An arbitrary vector, $|\psi\rangle$, belonging to the linear vector space can be expanded in this basis as

$$|\psi\rangle = \sum_i c_i |\phi_i\rangle,$$

where the expansion coefficients $c_i = \langle\phi_i|\psi\rangle$ are called the components of the vector ψ in the basis $\{|\phi_i\rangle\}$. Note that if $|\psi\rangle$ is normalized to unity, i.e., $\langle\psi|\psi\rangle = 1$ then

$$\langle\psi|\psi\rangle = \sum_i \sum_j \langle\phi_i|c_i^* c_j|\phi_j\rangle = \sum_i \sum_j c_i^* c_j \langle\phi_i|\phi_j\rangle$$

$$= \sum_i \sum_j c_i^* c_j \delta_{ij} = \sum_i |c_i|^2 = 1.$$

and

$$\sum_i |c_i|^2 = \sum_i c^* c_i = \sum_i \langle\psi|\phi_i\rangle\langle\phi_i|\psi\rangle = \langle\psi \left(\sum_i |\phi_i\rangle\langle\phi_i| \right) \psi\rangle = 1.$$

Completeness condition

$$\sum_i |\phi_i\rangle\langle\phi_i| = \hat{I}.$$

In the case of a continuous basis in which the vector functions depend on a continuous parameter α, the closure relation reads:

$$\int d\alpha\, |\phi(\alpha)\rangle\langle\phi(\alpha)| = \hat{I}.$$

Finally, let us note that in an orthonormal basis $\{\phi_i\}$,i = 1,2,3,...,n, an arbitrary ket, $|\psi\rangle$ (belonging to the vector space) is represented by a column matrix

$$|\psi\rangle = \begin{pmatrix} \langle\phi_1|\psi\rangle \\ \langle\phi_2|\psi\rangle \\ \langle\phi_3|\psi\rangle \\ . \\ . \\ . \\ \langle\phi_n|\psi\rangle \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ . \\ . \\ . \\ c_n \end{pmatrix}.$$

Schwarz inequality

For any two states $|\psi\rangle$ and $|\phi\rangle$ of the Hilbert space, we can show that

$$|\langle\psi \mid \phi\rangle|^2 \leq \langle\psi \mid \psi\rangle\langle\phi \mid \phi\rangle.$$

If $|\psi\rangle$ and $|\phi\rangle$ are linearly dependent (i.e., proportional: $|\psi\rangle = a|\phi\rangle$, where is a scalar), this relation becomes an equality.

The Schwarz inequality is analogous to the following relation of the real Euclidean space

$$|\vec{A} \cdot \vec{B}|^2 \leq |\vec{A}|^2 |\vec{B}|^2 .$$

Triangle inequality

$$\sqrt{\langle \psi + \phi \mid \psi + \phi \rangle} \leq \sqrt{\langle \psi \mid \psi \rangle} + \sqrt{\langle \phi \mid \phi \rangle}.$$

If $|\psi\rangle$ and $|\phi\rangle$ are linearly dependent and if the proportionality scalar is real and positive, the triangle inequality becomes an equality. The counterpart of this inequality in Euclidean space is given by

$$|\vec{A} + \vec{B}| \leq |\vec{A}| + |\vec{B}|.$$

$|\psi\rangle$ and $|\phi\rangle$ are said to be orthonormal if they are orthogonal and if each one of them has a unit norm:

$$\langle \psi \mid \phi \rangle = 0, \qquad \langle \psi \mid \psi \rangle = 1, \qquad \langle \phi \mid \phi \rangle = 1.$$

Observables and Operators

The measurable physical characteristics of a system, such as position, momentum, energy etc, are called observables and are represented by operators.

Mathematically, an operator, O, can be defined as a map $O : V \rightarrow V$ that takes a vector, belonging to a vector space V, to another vector also belonging to V.

In general, an operator is characterized by its action on the basis vectors of V and hence, in a chosen basis, it is represented by a matrix.

Observables and Operators

The action of an arbitrary operator A on a ket $|\psi\rangle \in V$ is written as:

$$\hat{A}\,|\psi\rangle = |\phi\rangle\,.$$

The product of an operator A and a number (complex) a is an operator aA, which takes a vector $|\psi\rangle \in V$ into the vector $a(A\psi\rangle) \in V$:

$$(a\hat{A})|\psi\rangle = a(\hat{A}|\psi\rangle.$$

The sum, C, of two operators A and B is defined as

$$\hat{C}\,|\psi\rangle = (\hat{A}+\hat{B})\,|\psi\rangle = \hat{A}\,|\psi\rangle + \hat{B}\,|\psi\rangle\,.$$

The operators in quantum mechanics are linear.

$$\hat{A}(\alpha\,|\psi\rangle + \beta\,|\phi\rangle) = \alpha\hat{A}\,|\psi\rangle + \beta\hat{A}\,|\phi\rangle\,.$$

Observables and Operators

Consider an operator A acting in V

$$|\phi\rangle = \hat{A}|\psi\rangle.$$

Let us introduce an operator $A^\dagger$ which acts in dual space V^* by taking the bra $\langle\psi|$,

$$\langle\phi| = \langle\psi|\hat{A}^\dagger.$$

Therefore,

$$\langle\psi|\hat{A}^\dagger|\chi\rangle^* = \langle\chi|\hat{A}|\psi\rangle.$$

Outer product: The outer product between a ket and a bra is written as

$$|\psi\rangle\langle\phi|.$$

Unity operator: it leaves any ket unchanged

$$\hat{I} \mid \psi\rangle = \mid \psi\rangle.$$

The gradient operator:

$$\vec{\nabla}\psi(\vec{r}) = (\partial\psi(\vec{r})/\partial x)\vec{i} + (\partial\psi(\vec{r})/\partial y)\vec{j} + (\partial\psi(\vec{r})/\partial z)\vec{k}.$$

The linear momentum operator:

$$\vec{P}\psi(\vec{r}) = -i\hbar\vec{\nabla}\psi(\vec{r}).$$

The Laplacian operator:

$$\nabla^2\psi(\vec{r}) = \partial^2\psi(\vec{r})/\partial x^2 + \partial^2\psi(\vec{r})/\partial y^2 + \partial^2\psi(\vec{r})/\partial z^2.$$

The parity operator:

$$\hat{\mathcal{P}}\psi(\vec{r}) = \psi(-\vec{r}).$$

Products of operators

The product of two operators is generally not commutative:

$$\hat{A}\hat{B} \neq \hat{B}\hat{A}.$$

The product of operators is, however, associative:

$$\hat{A}\hat{B}\hat{C} = \hat{A}(\hat{B}\hat{C}) = (\hat{A}\hat{B})\hat{C}.$$

We may also write

$$\hat{A}^n \hat{A}^m = \hat{A}^{n+m}$$

and

$$\hat{A}\hat{B} \mid \psi\rangle = \hat{A}(\hat{B} \mid \psi\rangle).$$

The *expectation* or *mean* value of an operator A with respect to a state

$$\langle \hat{A} \rangle = \frac{\langle \psi \mid \hat{A} \mid \psi \rangle}{\langle \psi \mid \psi \rangle}.$$

Hermitian Adjoint

The Hermitian adjoint, or simply the adjoint, $A^\dagger$, of an operator A is defined by this relation:

$$\langle \psi \mid \hat{A}^\dagger \mid \phi \rangle = \langle \phi \mid \hat{A} \mid \psi \rangle^*.$$

To obtain the Hermitian adjoint of any expression, we must cyclically reverse the order of the factors and make three replacements:

Replace constants by their complex conjugates: $\alpha^\dagger = \alpha^*$.

Replace kets (bras) by the corresponding bras (kets):

$$(\mid \psi \rangle)^\dagger = \langle \psi \mid \text{and } (\langle \psi \mid)^\dagger = \mid \psi \rangle.$$

Replace operators by their adjoints.

Hermitian Adjoint

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Following these rules, we can write

$$
\begin{aligned}
(\hat{A}^\dagger)^\dagger &= \hat{A}, \\
(a\hat{A})^\dagger &= a^* \hat{A}^\dagger, \\
(\hat{A}^n)^\dagger &= (\hat{A}^\dagger)^n, \\
(\hat{A}+\hat{B}+\hat{C}+\hat{D})^\dagger &= \hat{A}^\dagger + \hat{B}^\dagger + \hat{C}^\dagger + \hat{D}^\dagger, \\
(\hat{A}\hat{B}\hat{C}\hat{D})^\dagger &= \hat{D}^\dagger \hat{C}^\dagger \hat{B}^\dagger \hat{A}^\dagger, \\
(\hat{A}\hat{B}\hat{C}\hat{D} \mid \psi\rangle)^\dagger &= \langle\psi \mid D^\dagger C^\dagger B^\dagger A^\dagger.
\end{aligned}
$$

The Hermitian adjoint of the outer product is given by

$$(\mid \psi\rangle\langle\phi \mid)^\dagger = \mid \phi\rangle\langle\psi \mid .$$

Hermitian Adjoint

Operators act inside kets and bras, respectively, as follows:

$$|\,\alpha\hat{A}\psi\rangle = \alpha\hat{A}\,|\,\psi\rangle, \qquad \langle\alpha\hat{A}\psi\,| = \alpha^*\langle\psi\,|\,\hat{A}^\dagger.$$

and

$$\langle\psi\,|\,\hat{A}\,|\,\phi\rangle = \langle\hat{A}^\dagger\psi\,|\,\phi\rangle = \langle\psi\,|\,\hat{A}\phi\rangle.$$

An operator A is said to be *Hermitian* if it is equal to its adjoint $A^\dagger$:

$$\boxed{\hat{A} = \hat{A}^\dagger \quad \text{or} \quad \langle\psi\,|\,\hat{A}\,|\,\phi\rangle = \langle\phi\,|\,\hat{A}\,|\,\psi\rangle^*.}$$

On the other hand, an operator B is said to be *skew-Hermitian* or *anti-Hermitian* if

$$\boxed{\hat{B}^\dagger = -\hat{B} \quad \text{or} \quad \langle\psi\,|\,\hat{B}\,|\,\phi\rangle = -\langle\phi\,|\,\hat{B}\,|\,\psi\rangle^*.}$$

The Projection Operator

An operator P is said to be a projection operator if it is hermitian and equal to its own square

$$\hat{P} = \hat{P}^{\dagger}, \quad \hat{P}^2 = \hat{P}.$$

Clearly, the unit operator I satisfies these properties and is an example of a projection operator.

Consider an operator, A, equal to the outer product of a ket and its corresponding bra:

$$\hat{A} = |\phi\rangle\langle\phi|.$$

By definition it acts on a ket $|\psi\rangle$ through the rule

$$\hat{A}|\psi\rangle = (|\phi\rangle\langle\phi|)\,|\psi\rangle = |\phi\rangle\,\langle\phi|\psi\rangle.$$

The Projection Operator

The claim is that if $|\phi\rangle$ is normalized to unity, the operator A is a projection operator.

Consider the sum of two projection operators P_1 and P_2.

$$\hat{P}^\dagger = (\hat{P}_1\hat{P}_2)^\dagger = \hat{P}_2^\dagger\hat{P}_1^\dagger = \hat{P}_2\hat{P}_1,$$

$$\hat{P}^2 = (\hat{P}_1\hat{P}_2)^2 = \hat{P}_1\hat{P}_2\hat{P}_1\hat{P}_2 = \hat{P}_1(\hat{P}_2\hat{P}_1)\hat{P}_2.$$

It is quite clear from the aforementioned equations that P will satisfy the required properties for being a projection operator only if P_1 and P_2 commute. It is also clear that, if P_1 and P_2 commute, P does satisfy the required properties for being a projection operator.

Commutator Algebra

The *commutator* of two operators A and B, denoted by $[A, B]$, is defined by

$$[\hat{A},\ \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A},$$

and the anticommutator {A, B} is defined by

$$\{\hat{A},\ \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}.$$

Two operators are said to commute if their commutator is equal to zero and hence $AB=BA$. Any operator commutes with itself:

$$[\hat{A},\ \hat{A}] = 0.$$

Note that if two operators are Hermitian and their product is also Hermitian, these operators commute

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger = \hat{B}\hat{A},$$